

Individualised Strength

How MicroPulse turns your chosen exercises, each player's data, and the match countdown into one daily session — and how you keep control of every part of it.

For coaches + S&C

Coach → Strength

~20-min micro-dose

01 – The one idea

One session a day. You choose how it's built.

Every player gets a single strength session. You pick whether it's the standard team template you set up yourself, or an individualised version that adds each player's own layer — built from the exercises *you* chose for the team.

MODE A

Standard

The matchday template (MD-4 heavy, MD-3 power ...), tuned to the day and the player's readiness. The classic session — you enter it.

MODE B · TEAM DEFAULT

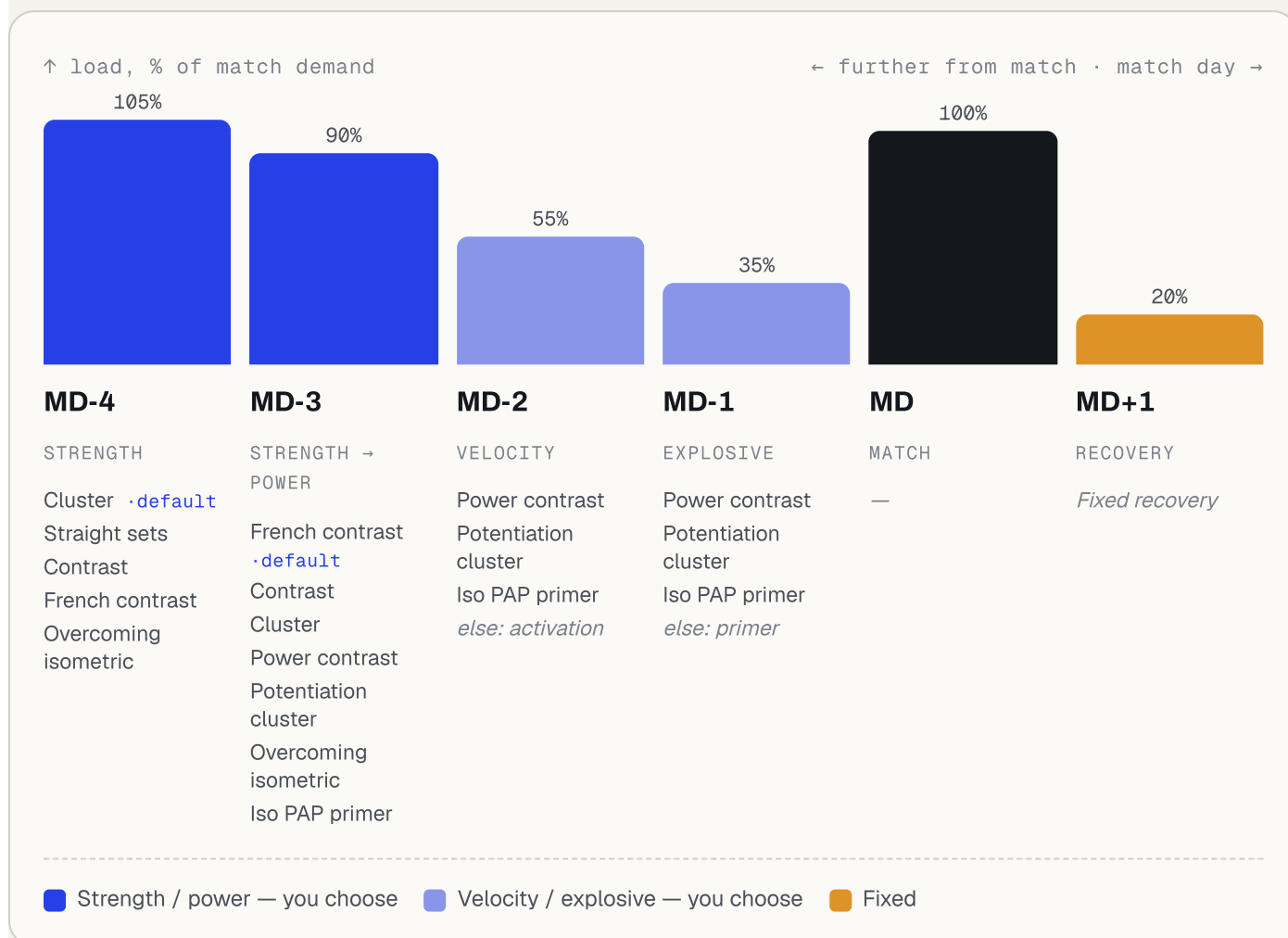
Individualised

The same template *plus* the player's own layer: their movement-screen corrective, their strength/power deficits, and their left/right symmetry — from your palette.

Why one session? In-season, 15–20 minute micro-doses 3–5×/week preserve strength and power without next-day fatigue. [Rønnestad 2023](#).

Pick the method per day. The load already follows the countdown.

You tie a training *method* to each day; MicroPulse lays it out from your palette. The days near the match only offer light, fast methods — the intent there is speed, not load. Heavy methods are locked to MD-4 / MD-3 on purpose.



Bars show each day's load as % of match demand — the taper the periodisation already applies. [Martin-García 2018](#). Leave a day on *Default* to keep its built-in structure; injury-prevention work (Nordic, Copenhagen) always stays on the strength days.

One method per day — and it adapts. You don't stack methods (that bloats the micro-dose). Instead the chosen method **steps down a rung for a yellow player** — French contrast → Contrast, Contrast → Cluster. Green keeps it, red is already recovery, and the set-reduction still applies on top. Two team-mates, same day: the recovered one gets the complex, the tired one a lighter version.

The family ladder

The downgrade stays **within the training quality** — a strength day steps to a lighter strength/isometric method, a velocity day to a lighter velocity method, never crossing between them.

STRENGTH FAMILY — HEAVIEST → LIGHTEST

French contrast

- ↓ Contrast
- ↓ Cluster
- ↓ Overcoming isometric
- ↓ Straight sets

VELOCITY FAMILY — HEAVIEST → LIGHTEST

Power contrast · or Potentiation cluster

- ↓ Iso PAP primer

A yellow player steps **exactly one rung down**; green keeps the chosen method; red is recovery. In the app the coach previews this per day with the “*show ladder*” link in the structure picker — the method they picked is marked green, the yellow landing amber.

Pick the pool. The system picks per player.

The individualised (and auto) build assembles each player's work from a pool you choose. Pick **both** a bilateral and a unilateral lower-body option — the system chooses between them per player from their left/right symmetry.

Power / explosive

Jumps · olympic · ballistic · med-ball

Bilateral strength

Squat · trap-bar · hip thrust

Unilateral strength

Split squat · B-stance RDL

Isometric · strength / RFD / PAP

IMTP · iso squat 90° · Spanish squat

Symmetry decides the lower-body lift. A player $\geq 10\%$ left/right asymmetry gets the single-leg lift to load the weaker side; a symmetric player gets the bilateral lift. [Bishop 2020](#). Leave a slot empty → the built-in default. The palette never touches the Standard session.

Isometrics — for strength *and* power. Two methods you can tie to a day: **Overcoming isometric** (strength days) — push maximally against an immovable bar at a deep joint angle for max strength + rate of force, joint-friendly; and the **Isometric PAP primer** (taper days) — a maximal isometric conditioning set that potentiates the explosive work 3–6 min later. [Oranchuk 2023](#) · [Krzysztofik 2023](#) · [Jarosz 2025](#).

Swap, but only for a safe alternative.

When the sent exercise isn't quite right — equipment, comfort, a niggle — both sides can swap it, always from that exercise's curated safe alternatives.

YOU · THE COACH

On a player's session, hit  **swap**. Safe alternatives are marked and listed first, then the rest of the same category.

THE PLAYER

A “**Swap an exercise (safe options)**” panel on their Today card — safe alternatives only, never anything inappropriate. Every player swap is logged for you.

Let it send every morning — if you want.

MicroPulse can send each player their session automatically on strength days, no click. It's self-service per team and  **off by default**

- **You turn it on/off** for your own team on Coach → Strength.
- **Never overwrites** a session you already sent that day.
- **Skips** off-days, injured players, and anyone on a RECOVERY verdict.
- You can always re-send, override, or switch it off.

Rules decide. The system explains. You have the last word.

- **Every change shows its “why”** and a paper citation under “*Why these changes?*” — a symmetry-driven unilateral lift, a chosen method, a corrective.
- **Your manual swap overrides any system choice** — the engine recommends, you decide.
- **Nothing is hidden:** mode, palette, structures and auto-send are all your explicit settings. The player sees one session, labelled “Sent by your coach”, tuned to how they are today.

Under the hood

For S&C staff and developers — the pipeline, thresholds, data model and evidence behind the coach-facing behaviour above.

Build pipeline

`buildStrengthSession(snapshot, coachOverrides, { mode })` — one player, one day:

- 1 **Template / structure** — default → `pickTemplate(md)`; a chosen non-default allowed method → `buildStructuredBlocks(key, md)`.
- 2 **Adaptation rules** — readiness, soreness, decel burden, VBT, sprint drop, CoD asymmetry, congestion. Standard mode drops the deficit-ledger emphases.
- 3 **Team palette** (individualised only) — substitutes your exercises into the power/strength slots at the MD dose; symmetry picks the pool.
- 4 **Coach overrides** — your manual swaps win, last.
- 5 **Correctives** — the movement-screen corrective merges as the front activation block, not readiness-reduced.
- 6 **Serialize** → the Today-card structure, one row per player per day.

Symmetry & structures

- `codAsymmetryPct ≥ 10%` (Bishop 2020) → unilateral pool, else bilateral. **No cross-pool fallback** — an empty pool leaves the template exercise, so the palette never undoes the engine's own CoD-asymmetry swap.
- Explosive taper methods force a light + fast dose (10% velocity cap) regardless of the day and **skip** the Nordic/Copenhagen block; strength methods keep it. Prevention is never palette-driven.
- Dose is exercise+MD specific and sparse, so a method landing on an undosed day resolves to the nearest MD dose.
- **Readiness downgrade** — a **MODIFIED** / **REDUCED** verdict steps the method one rung down a demand ladder (`STRUCTURE_DEMAND`) to the highest lower-demand method allowed that day; applies to the day's default too. Audit `STRUCTURE_READINESS_DOWNGRADE`. Set-reduction still stacks on top.

Data model

STORE	HOLDS	ACCESS
<code>teams.strength_send_mode</code>	team default: individualised · standard	coach
<code>teams.strength_auto_send</code>	opt-in morning send (bool)	coach
<code>team_strength_palette.slots</code>	per-slot exercise pool (jsonb)	RLS <code>coach_team_ids()</code> / <code>is_staff()</code>
<code>team_strength_palette.md_structures</code>	MD → method map (jsonb)	same
<code>player_today_strength_override</code>	the sent session · 1 row/player/date	coach-written
<code>strength_session_overrides</code>	coach manual swaps	coach, team-scoped
<code>player_exercise_swaps</code>	player swaps (log + coach window)	player-owned · coach read

Endpoints

ROUTE	DOES
GET/POST /api/coach/team/strength-palette	palette slots + MD structures (POST merges only the field sent)
GET/POST /api/coach/team/strength-send-mode	team default mode + auto-send
POST /api/coach/team/send-strength-sessions	bulk send (per-send mode override)
POST /api/coach/player/[id]/send-strength-session	single send
POST /api/coach/player/[id]/strength-override	coach swap — safe alternatives first
POST/DELETE /api/player/exercise-swap	player swap — validates target ∈ alternatives, logs, rewrites today
GET /api/strength/auto-send	Vercel cron 0 6 * * * · opt-in · skip-if-sent · strength days

Audit rules — surfaced as “Why these changes?”

STANDARD_MODE	TEAM_PALETTE_APPLIED	PALETTE_SYMMETRY_UNILATERAL
STRUCTURE_APPLIED	CORRECTIVE_MERGED	CORRECTIVE_CORROBORATES_EMPHASIS
COACH_OVERRIDE_APPLIED	INJURY_BLOCK	

Evidence base

Rønnestad 2023 microdosing · Bishop 2020 inter-limb asymmetry · Tufano 2017 cluster sets · Cormie 2011 / Liu 2023 contrast & French contrast · Pareja-Blanco 2017 velocity-loss caps · van Dyk 2019 Nordic (–51% hamstring) · Harøy 2019 Copenhagen (–41% groin) · Martin-García 2018 matchday taper · Oranchuk 2023 isometric length/intensity/intent · Krzysztofik 2023 / Jarosz 2025 isometric conditioning → PAPE · Schaefer & Bittmann 2017 pushing vs holding.